Confluent Kafka Isotope

`confluent-kafka-isotope` is a reference implementation of an e-commerce order pipeline that uses Kafka Interceptors with Apache Flink to capture and report event tracing data — both in batch and near real time.

Much like isotopes traced through a biochemical pathway, each event carries metadata that allows it to be tracked as it moves through Kafka topics and distributed microservices.

Kafka topics become the connective tissue between services, while Kafka Interceptors quietly transform the pipeline itself into an observable distributed system.

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1.0 How the isotope is carried

This project renders a Kafka pipeline as a **bipartite graph** — services on one vertex set, topics on the other, edges crossing in both directions — so every produce edge, every consume edge, and every terminal consumer is a first-class node in the same topology view. The mechanism is a tracer (an *isotope*) carried in record headers, captured by two library calls: a Kafka **producer interceptor** stamps an isotope and appends one hop per `send()` (produce edges → `hops[]`), and `IsotopeContext.recordConsume(record, service, producer)` forwards that isotope to a value-less marker on `platform.observability.consume_events` (consume edges). Consume-then-produce services additionally call `IsotopeContext.adoptFromRecord(record)` between consume and produce so the trace ID survives each hop. Apache Flink reads only the headers and reports on what the isotopes reveal: **end-to-end latency**, the **full bipartite service↔topic↔service graph**, **drop/duplication rates**, and **pipeline coverage**.

A **portability requirement** runs through this project: the same isotope mechanism must work against both **Confluent Cloud for Apache Flink (CCAF)** (managed) and **Confluent Platform for Apache Flink** (self-managed) — both used here via Table API SQL plus uploaded UDF/PTF JARs (no DataStream code on either side). Three decisions follow from that: tagging happens in a Kafka **producer interceptor** (the one extension point both runtimes share via the broker), the on-wire **header** format is **JSON** (so Flink SQL can read the scalar fields with `CAST(headers[…] AS STRING)` and no

UDF), and the optional stateful reports (`LatencyPercentilesUDAF`, `StuckTracePTF`) ship as a single JAR that registers identically on either runtime.

One asymmetry the runtimes don't share: **Flink-native SR-Protobuf**. CCAF supports SR-framed Protobuf as a Flink sink format via its topic catalog; Apache Flink open-source (the CP Flink runtime) ships `avro-confluent` but no SR-Protobuf counterpart. So Flink *report sinks* land on **Avro+SR on CP** and can be **Protobuf+SR on CCAF** — a runtime constraint, not a project preference. The demo *event* topics (next paragraph) are unaffected because they're written by the Kafka producer client, not by Flink.

Message **values** on the demo topics are **SR-framed Protobuf** (`ai.signalroom.kafka.isotope.proto.DemoEvent`) — the standard Confluent value format. The interceptors and reports are agnostic to value format because the isotope rides in headers; the Protobuf choice just gives the integration tests and the demo CLI a typed payload to work with. The **headers** are where the isotope lives, and they have two parts:

- **Header `x-isotope`** (JSON bytes) carries the full hop history, forwarded by every hop:
  - `t` — 16-byte **UUIDv7** trace ID (RFC 9562): 48-bit ms timestamp in the high bits + 74 bits random. Stable for the life of the trace, and lexicographic byte order matches creation order — sort trace IDs and you get chronological order for free. The random bits come from `ThreadLocalRandom`, not `SecureRandom`: a trace ID is a public observability identifier carried in Kafka headers, so the requirement is collision avoidance, not unpredictability — and `ThreadLocalRandom` delivers that without `SecureRandom`'s per-call cost on every produced record.
  - `o` — origin timestamp (ms) — same value as the timestamp embedded in the UUIDv7 trace ID; kept as its own field for typed access from Flink SQL without needing to decode the UUID bytes.
  - `s` — origin service name (set once, never reassigned)
  - `p` — origin pipeline name (e.g. `orders` vs `location`); like `s`, stamped once at the origin and forwarded unchanged on every hop, so reports can slice traces by which logical pipeline they belong to
  - `h` — ordered list of hops, each `{s: service, t: topic, m: tsMs}`
  - `x` — `true` if the hop list exceeded `MAX_HOPS = 32` and the oldest hop was evicted
- **Seven scalar headers** (UTF-8 strings) carry the most-recent-hop view so Flink SQL can read them via `CAST(headers['x-isotope-…'] AS STRING)` without parsing the JSON array (no UDF required on either CCAF or CP Flink). See scripts/flink/README.md for the full header table.

A producer with the isotope interceptor loaded appends one hop on every `send()`. A consume-then-produce service calls `IsotopeContext.adoptFromRecord(record)` between consume and produce so the trace ID and origin survive the hop.

1.1 How the producer interceptor gets invoked

Application code never calls `onSend` / `onAcknowledgement` directly — the Kafka client invokes them by reflection once two things are in place:

1. **Register the class in producer config.** Put `IsotopeProducerInterceptor.class.getName()` under `interceptor.classes` (App.java:199, App.java:284, IsotopeTestHarness.java:96). The `KafkaProducer` constructor instantiates the interceptor and owns its lifecycle.
2. **Call `producer.send(...)` as normal.** Every `send()` triggers `onSend` (caller thread, before serialization) and later `onAcknowledgement` (producer I/O thread, after broker ack or failure).

The exact call sites in `kafka-clients` 4.2.0: `KafkaProducer.send` invokes `interceptors.onSend` (line 950) and `AppendCallbacks.onCompletion` invokes `interceptors.onAcknowledgement` (line 1600). Exceptions thrown by the interceptor are caught and logged by the client's fan-out wrapper — they never break the `send` call.

1.2 Using the Kafka client interceptor vs explicit library calls

This project uses one Kafka client interceptor, and it's a `ProducerInterceptor` — not a `ConsumerInterceptor`. A `ConsumerInterceptor.onConsume` sees a whole batch from `poll()`, but the application processes records one at a time, so a thread-local snapshot from the batch would be ambiguous about which record is being handled. An earlier

`IsotopeConsumerInterceptor` was tried and removed for exactly that reason (see #30). The consume side instead runs on **explicit per-record library calls**: services call `IsotopeContext.adoptFromRecord(record)` between consume and produce to carry the trace through a hop (the next `send()` then sees that thread-local and appends a new hop), and `IsotopeContext.recordConsume(record, service, markerProducer)` to emit a consume-edge marker (see the next paragraph). So: **producer interceptor for produce-side stamping; explicit per-record calls for everything consume-side**.

1.3 Why explicit calls for consume-side markers?

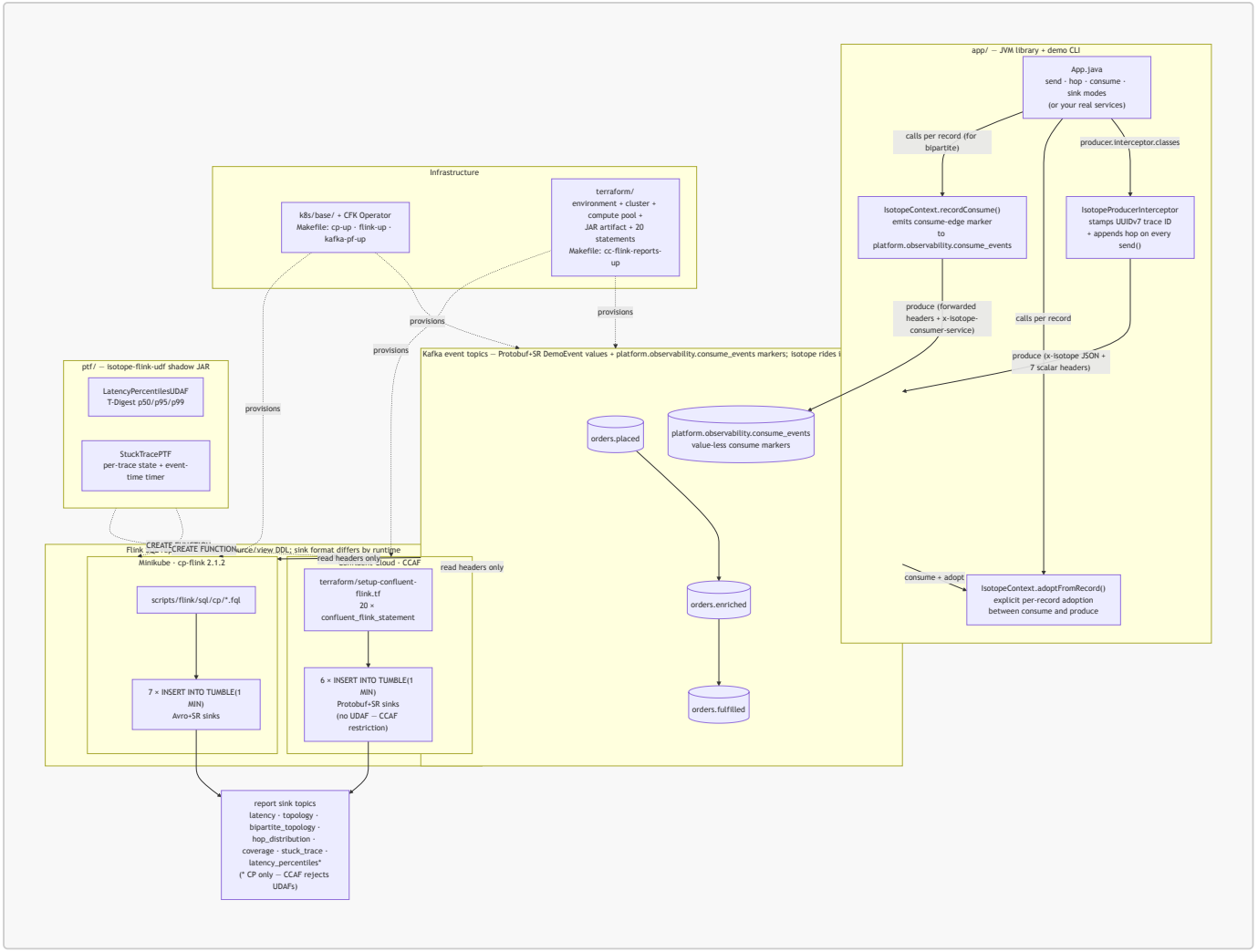
Consume-side markers — the other half of the bipartite graph. The produce-side `hops[]` chain captures `producer → topic` edges and implicitly captures consume edges for *intermediate* services (svc-B consuming from topic-AB is implied by svc-B's next produced hop), but **terminal consumers** — services that consume but never produce — would otherwise be invisible. The bipartite story closes that gap with one library call: consumers call `IsotopeContext.recordConsume(record, service, markerProducer)` between consume and process, which forwards the inbound record's seven scalar `x-isotope-*` headers to a value-less marker record on `platform.observability.consume_events` and adds one new header — `x-isotope-consumer-service` — that names the consumer. The `bipartite_topology` Flink report unions these markers with the produce-side view to emit edges in both directions: `producer → topic` (from `isotope`) and `topic → consumer` (from `consume_events`). Markers are fire-and-forget: a dropped marker leaves a hole in the topology graph but never disrupts the consume/produce pipeline.

1.4 Why "bipartite"?

Background — why "bipartite"? The resulting topology report is an example of a **bipartite graph** from graph theory (a sub-field of discrete mathematics). A graph is *bipartite* when its vertices partition into two disjoint sets such that every edge connects a vertex in one set to a vertex in the other — edges never run within a set. Here the two sets are **services** (`svc-A`, `svc-B`, …) and **topics** (`orders.placed`, `orders.enriched`, `orders.fulfilled`, `platform.observability.consume_events`); every edge is either a **produce edge** (`service → topic`, from the isotope's `hops[]`) or a **consume edge** (`topic → service`, from the `platform.observability.consume_events` markers). Kafka's pub/sub model guarantees the bipartite shape: services never connect directly to other services, and topics never connect directly to other topics — every interaction flows through the opposite set. Before consume-side markers existed, only produce edges were captured, so the topology view collapsed into a **unipartite** `service → service → service` chain with topics hidden as edge labels and terminal consumers omitted entirely. With both edge directions now wired, topics become first-class nodes alongside services, and the `bipartite_topology` report renders the full graph — every produce *and* consume edge, in both directions.

2.0 Architecture

A bird's-eye view of the moving parts. The JVM library in app/ registers a Kafka producer interceptor that stamps the isotope into record headers on every `send()`; consume-then-produce services adopt the inbound trace via an explicit `IsotopeContext.adoptFromRecord(record)` call; records flow through a 3-topic chain; Flink SQL reads only the headers and emits 1-minute aggregate reports. The same source/view DDL deploys to both runtimes — **CP** on Minikube applies `.fql` files under scripts/flink/sql/cp/, and **CCAF** in Confluent Cloud applies inline `confluent_flink_statement` resources under terraform/. The shadow JAR from ptf/ (which powers two of the six reports) registers identically on both. (Kafka is drawn once below for brevity — each runtime provisions its own cluster.)



See § 3.0 for the file tree behind each box, and § 4.0 for the run commands.

3.0 Repo layout

```
app/                                        isotope JVM library + demo CLI + tests
  src/main/proto/ai/signalroom/kafka/isotope/proto/
    demo_event.proto                        DemoEvent message (Protobuf value schema)
  src/main/java/ai/signalroom/kafka/isotope/
    Isotope.java                            POJO + JSON codec + Hop + fromHeaders()
                                            + UUIDv7 helpers (uuidV7Bytes / uuidV7String)
    IsotopeContext.java                     ThreadLocal + adoptFromRecord() +
                                            recordConsume() (emits consume-edge markers)
    IsotopeProducerInterceptor.java         stamps/appends x-isotope + 7 scalar
                                            reporting headers on send()
    App.java                                demo CLI — send / hop / consume / sink modes
  src/test/java/.../                        IsotopeCodecTest,
IsotopeContextRecordConsumeTest
                                            (no broker needed)
  src/integrationTest/java/.../             BrokerSmokeIT, ProducerInterceptorIT,
                                            ThreeStageHopPropagationIT,
BipartiteTopologyIT,
                                            IsotopeTestHarness — live-broker tests;
produce/consume
                                            DemoEvent via SR-framed Protobuf
                                            (need Minikube CP + SR port-forwarded)
ptf/                                        Flink PTF + UDAF shadow JAR (powers 2 of 6
reports)
  src/main/java/ai/signalroom/kafka/isotope/flink/
```

```
    LatencyPercentilesUDAF.java        T-Digest p50/p95/p99 aggregate
    StuckTracePTF.java                 per-trace state + event-time timer
    Percentiles.java                   p50/p95/p99 return-type DTO
  src/test/java/.../                   LatencyPercentilesUDAFTest
k8s/base/                              CFK manifests
  confluent-platform-c3++.yaml         Kafka / SR / Connect / ksqlDB / Control Center
  flink-basic-deployment.yaml          cp-flink session cluster + CMF
  flink-rbac.yaml                      RBAC for the cp-flink operator
scripts/
  port-forward-kafka.sh                localhost:30092 → Kafka, localhost:8081 → SR
  port-forward-taskmanager.sh          Flink TaskManager web UI forward
  deploy-cp-flink-reports.sh           builds shadow JAR + applies sql/cp/*.fql to
                                       the cp-flink session cluster
  deploy-cc-flink-reports.sh           builds shadow JAR + wraps `terraform apply`
                                       for the CCAF path
  cc-cli-env.sh                        pulls Kafka + SR creds from `terraform
output`,
                                       builds the JAAS string, exports BOOTSTRAP /
                                       SR_URL / KAFKA_KEY / KAFKA_SECRET / JAAS / ...
  cc-app-run.sh                        thin wrapper around `./gradlew :app:run` that
                                       sources cc-cli-env.sh and injects the six -D
flags
  flink/README.md                      Flink SQL reports — runtime split (CP=7
reports/Avro+SR,
                                       CCAF=6 reports/Protobuf+SR), layout,
operations
  flink/sql/cp/                        CP Flink SQL: 00_source_table,
01_register_functions,
                                       05_isotope_view, 06_consume_events_view,
                                       05_report_sinks (avro-confluent),
                                       10/20/25/30/40/60/70 INSERT INTO reports,
99_teardown
                                       (CCAF SQL is inlined under terraform/setup-
confluent-flink.tf.)
terraform/                             CCAF infrastructure-as-code (`make cc-flink-
reports-up`)
  providers.tf                         Confluent provider — cloud key/secret vars
  versions.tf                          required Terraform (>= 1.13) + provider
versions
  variables.tf                         confluent_api_key/secret, cloud, region,
day_count
  data.tf                              organization lookup + other data sources
  setup-confluent-environment.tf       environment (ESSENTIALS stream-governance
package)
  setup-confluent-kafka.tf             Kafka cluster + Kafka API key rotation module
                                       (iac-confluent-api_key_rotation-tf_module)
  setup-confluent-flink.tf             service account + 6 role bindings, compute
pool,
                                       artifact upload, SR API key rotation, and 20
inline
                                       `confluent_flink_statement` resources: 4 ALTER
TABLE
                                       + 3 VIEW + 6 sink CREATE TABLE + 1 CREATE
FUNCTION
                                       (PTF; UDAF skipped — CCAF rejects UDAFs) + 6
INSERT INTO
  outputs.tf                           environment_id, bootstrap, SR URL, rotating
                                       Kafka + SR API key/secret outputs (sensitive)
  terraform.png                        rendered resource graph (embedded in § 4.5)
Makefile                               cp-up / flink-up / kafka-pf-up / flink-
reports-up /
```

```
                                        cc-flink-reports-up / cc-flink-reports-down /
...
```

4.0 Running

Cheapest-first order if you're new: `./gradlew test` (§ 4.1) → local CP via Minikube (§ 4.2–4.4) → CCAF in the cloud (§ 4.5). Skip ahead if you only care about one runtime.

4.1. Unit tests (no broker, instant)

```
./gradlew test                        # both subprojects
# or scoped:
./gradlew :app:test                   # IsotopeCodecTest (10) +
IsotopeContextRecordConsumeTest (4) — JSON roundtrip, hop eviction, UUIDv7 properties,
consume-marker emission (14 tests)
./gradlew :ptf:test                   # LatencyPercentilesUDAFTest — T-Digest
accumulator semantics
```

4.2 Demo CLI — see one trace propagate live

The fastest way to watch the isotope mechanic. Requires the cluster to be up and the Kafka + SR forwards running (see step 3 below for the bring-up commands). The CLI has two argument styles — **pipeline-position verbs** that bake in the orders.* topic chain (recommended for the demo) and **generic verbs** that take raw topic + service args (for ad-hoc inspection on any topic):

Verb	Args	What it does
`place`	`[payload]`	Produces one isotope-tagged `DemoEvent` to `orders.placed` as `order-intake-service` (default payload: `hello`), then exits. Auto-creates the topic.
`enrich`	—	Consumes from `orders.placed`, adopts the isotope, **emits a consume-edge marker to `platform.observability.consume_events` as `order-enrichment-service`**, then re-produces to `orders.enriched`. Runs until Ctrl-C.
`fulfill`	—	Same as `enrich` but for `orders.enriched → orders.fulfilled` as `order-fulfillment-service`. Runs until Ctrl-C.
`ship`	—	Terminal consumer for `orders.fulfilled` as `shipping-notification-service`. **Emits a consume-edge marker** so it shows up in the bipartite report; does not re-produce. Runs until Ctrl-C.
`send`	`<topic>` `<service>` `<payload>`	Generic produce. Auto-creates the topic.
`hop`	`<in-topic>` `<out-topic>` `<service>`	Generic consume-then-produce; emits a consume-edge marker. Runs until Ctrl-C.
`consume`	`<topic>` `<service>`	Generic terminal-consume; emits a consume-edge marker and pretty-prints the trail. Runs until Ctrl-C.
`sink`	`<topic>`	Passive peek — pretty-prints the isotope trail but does NOT emit a consume marker. Use for ad-hoc inspection. Runs until Ctrl-C.

4.2.1 Full Bipartite Demo

A 4-stage chain (full bipartite graph) in four terminals. Run them in pipeline order — `place` produces, then `enrich` / `fulfill` / `ship` each pick up where the previous stage left off:

```
# Terminal A — kick the chain off (run repeatedly to send more)
./gradlew :app:run --args="place 'hello world'" -q

# Terminal B — first hop: orders.placed → orders.enriched
./gradlew :app:run --args="enrich" -q

# Terminal C — middle hop: orders.enriched → orders.fulfilled
./gradlew :app:run --args="fulfill" -q

# Terminal D — terminal consumer (prints the full 3-hop trail AND emits a
#              consume-edge marker so shipping-notification-service shows up
#              in the bipartite report)
./gradlew :app:run --args="ship" -q
```

Terminal D's output for each record shows the same `trace_id` across all three hops, `origin = order-intake-service` (never reassigned), and `hops[]` listing `order-intake-service → order-enrichment-service → order-fulfillment-service` in order with per-hop timestamps. The bipartite-topology report sees all six edges: produce edges `order-intake-service → orders.placed`, `order-enrichment-service → orders.enriched`, `order-fulfillment-service → orders.fulfilled` and consume edges `orders.placed → order-enrichment-service`, `orders.enriched → order-fulfillment-service`, `orders.fulfilled → shipping-notification-service`. Swap `consume` for `sink` if you only want to inspect records without recording the terminal edge. Override endpoints via `-Dkafka.bootstrap=…` / `-Dschema.registry.url=…` if you're not on the default Minikube layout.

4.3 Integration tests (live Kafka via Minikube)

Bring up the local Confluent Platform stack and port-forward Kafka + SR:

```
make minikube-start                 # one-time
make cp-up                          # CFK Operator + Kafka/SR/Connect/ksqlDB/C3 (~5
min)
make kafka-pf-up                    # localhost:30092 → Kafka, localhost:8081 →
Schema Registry
```

Then run the suite:

```
./gradlew :app:integrationTest                                          # every IT
below
./gradlew :app:integrationTest --tests '*ProducerInterceptorIT'         # just one
```

Override the endpoints if needed:

```
./gradlew :app:integrationTest \
    -PkafkaBootstrap=localhost:30092 \
    -PschemaRegistryUrl=http://localhost:8081
```

Tear down forwards when done:

```
make kafka-pf-down
```

The integration tests cover:

Test	What it verifies
`BrokerSmokeIT`	AdminClient can create/list/delete a topic via the NodePort port-forward
`ProducerInterceptorIT`	A consumer sees the `x-isotope` JSON header + all 7 scalar reporting headers with the expected origin/hop values, and the Protobuf round-trip preserves `DemoEvent.source` / `payload`
`ThreeStageHopPropagationIT`	`order-intake-service → topic-AB → order-enrichment-service → topic-BC → order-fulfillment-service` produces a stable trace ID, 2-hop trail in send order, and correct scalar headers (origin = `order-intake-service`, this = `order-enrichment-service`, hop count = 2) at the terminal; consume-then-produce hops use `IsotopeContext.adoptFromRecord` to carry the trace forward
`BipartiteTopologyIT`	The 4-stage `order-intake-service → topic-AB → order-enrichment-service → topic-BC → order-fulfillment-service → topic-CD → shipping-notification-service` chain emits exactly three consume-edge markers to a per-test markers topic — one per consume edge. Every marker carries the trace ID, forwarded `x-isotope-*` scalars describing the upstream producer, and the new `x-isotope-consumer-service` naming the downstream consumer. Asserts the `(consumer_service, consumed_topic)` set is exactly the three pairs of stages 2-4

4.4 Flink SQL reports on Confluent Platform for Apache Flink (Minikube)

The five pure-SQL reports plus the two JAR-backed reports (one PTF, one UDAF) — seven in total — run against a Flink session cluster managed by the Confluent Flink Kubernetes Operator. Same FQL files deploy to Confluent Cloud for Apache Flink — see **§ 4.5** for that path; this section is the local-Minikube path.

Bring up Flink:

```
make flink-up                # cert-manager → CFK Flink Operator → CMF → session
cluster
                             # (~5 min the first time)
```

Deploy the reports — all 7 reports run on the cp-flink session cluster (Flink 2.1.2). Sink topics use Apache Flink's `avro-confluent` format — SR-framed Avro, auto-registered on first write — so Control Center renders the report rows natively.

```
make flink-reports-up
```

`flink-reports-up` builds the PTF/UDAF shadow JAR if missing, copies it into the JobManager pod, pre-creates the 7 sink Kafka topics, then applies the source + view + sink DDL and submits 7 `INSERT INTO` streaming jobs (one per report). On first write to each sink, Apache Flink's `flink-sql-avro-confluent-registry` format registers a fresh Avro schema in SR under subject `<topic>-value`. **Control Center deserializes all 7 report topics natively** — no `.proto` files in the repo for the reports, no hand-installed format jars beyond the one init-container download.

4.4.1 Sustained traffic — required to see report rows.

All seven INSERT INTO jobs aggregate over `TUMBLE(event_time, INTERVAL '1' MINUTE)` windows, and a tumbling window only emits when the watermark advances past `window_end`. A handful of records bursted within a single 1-minute interval will sit in one open window forever (the most-recent record is the watermark, and it never gets older than itself). Spread traffic across **multiple** windows so the watermark crosses each boundary — same approach as § 4.5 but driving the local Minikube stack directly via gradle (no wrapper script needed — `App.java`'s defaults already point at `localhost:30092` / `localhost:8081`):

```
# Prereq: 'make kafka-pf-up' must be running so the port-forwards are live.
# 30 records spaced 5s apart ≈ 2.5 minutes of event-time → spans 3+ windows
for i in {1..30}; do
  ./gradlew :app:run --args="place burst-$i" -q
  sleep 5
done
```

Wait ~90 seconds after the *last* record before checking `isotope_report_latency_1m` (and friends) — that's the watermark catching up. The `stuck_trace_alerts_1m` sink only fires for traces that go ≥60s of event time without a fresh hop, so the burst above won't trigger it (every trace gets one record and ends — no stalled in-flight state). To exercise `STUCK_TRACE_PTF`: send a single record to `orders.placed` and don't run the `order-enrichment-service` / `order-fulfillment-service` hops, then keep sending unrelated records elsewhere so the watermark advances past the stuck trace's `event_time + 60s`.

4.4.2 Format-by-domain

The demo *event* topics (`orders.placed`, `orders.enriched`, `orders.fulfilled`) still ride **Protobuf+SR** via the Java app's `DemoEvent` schema — that's unchanged. The consume-edge marker topic `platform.observability.consume_events` has **null value + scalar headers only** (Flink reads the headers via a `MAP<STRING, BYTES>` virtual column; there's nothing to deserialize). The *report* topics ride **Avro+SR** because cp-flink doesn't ship an SR-integrated Protobuf format and CMF (which does) disallows the UDAFs the percentiles report needs. Events from the app are Protobuf; aggregates from Flink are Avro; consume markers are headers-only. Format by domain — a clean split, not a defect.

4.5 Flink SQL reports on Confluent Cloud for Apache Flink (CCAF)

CCAF parallel of § 4.4, driven by Terraform under terraform/. One `make` target spins up a fresh Confluent Cloud environment, Kafka cluster, 4 pre-created event topics (the three demo topics `orders.placed` / `orders.enriched` / `orders.fulfilled` plus the consume-edge marker topic `platform.observability.consume_events`; report sink topics are created on the fly by the Flink `CREATE TABLE` statements — see the comment in terraform/setup-confluent-kafka.tf for why pre-creating them via `confluent_kafka_topic` would conflict with CCAF's Topic Catalog auto-import), a Flink compute pool, two rotating service-account API key pairs (one for Kafka, one for Schema Registry), the PTF/UDAF JAR uploaded as a Flink artifact, and 20 long-lived `confluent_flink_statement` resources broken down as **4 ALTER TABLE** (add scalar headers on the event topics) + **3 VIEW** (raw + typed produce + typed consume) + **6 sink CREATE TABLE** + **1 CREATE FUNCTION** (PTF only — the UDAF is intentionally skipped, see the limitation note below) + **6 INSERT INTO** streaming jobs. The Terraform shape mirrors `apache_flink-kickstarter-ii` — same provider version, same `iac-confluent-api_key_rotation-tf_module`, same DROP-then-CREATE statement pattern.

4.5.1 Provisioning and Deployment commands

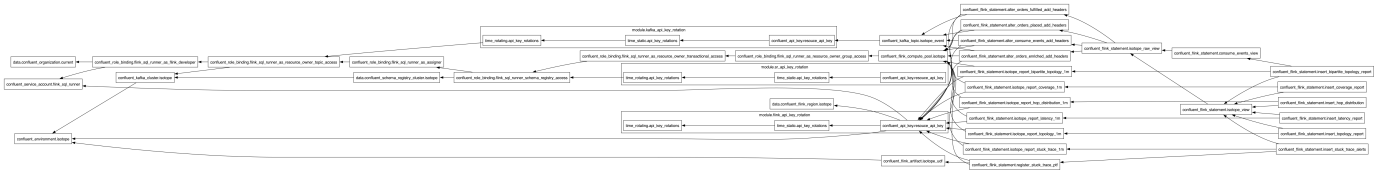
Prereqs:

- Terraform `>= 1.13` installed locally.
- A Confluent Cloud API key (Cloud-level, not cluster-scoped) with permissions to create environments, Kafka clusters, Flink compute pools, service accounts, role bindings, Flink artifacts, and statements. Generate via

Console → Settings → Cloud API keys.

Deploy:

```
export CONFLUENT_API_KEY=...
export CONFLUENT_API_SECRET=...
make cc-flink-reports-up CONFLUENT_API_KEY=$CONFLUENT_API_KEY
CONFLUENT_API_SECRET=$CONFLUENT_API_SECRET
```



The wrapper script (scripts/deploy-cc-flink-reports.sh) builds the PTF/UDAF shadow JAR if missing, then runs `terraform apply -auto-approve` in terraform/. First-run takes ~6–8 minutes (Kafka cluster provisioning dominates). Re-applies are idempotent — `CREATE … IF NOT EXISTS` plus `lifecycle { ignore_changes = [compute_pool] }` on every statement.

What gets created (see terraform/setup-confluent-flink.tf for the full graph):

Resource	Name	Notes
`confluent_environment`	`confluent-kafka-isotope`	ESSENTIALS stream-governance package
`confluent_kafka_cluster`	`kafka-isotope`	Standard, single-zone, AWS us-east-1 by default
`confluent_kafka_topic` × 4	`orders.{placed,enriched,fulfilled}` + `platform.observability.consume_events`	Only the event topics + the consume-marker topic are pre-created. The 6 `isotope_report_*_1m` sink topics are created on first deploy by their `CREATE TABLE` statement (CCAF's Topic Catalog auto-imports any pre-existing topic as `(key BYTES, val BYTES)`, which would silently no-op the typed `CREATE TABLE`). `terraform destroy` cleans them up via the environment cascade.
`confluent_service_account` + 6 role bindings	`isotope-flink-sql-runner`	FlinkDeveloper (org) + ResourceOwner on topic=* / transactional-id=* / group=* / SR subject=* + Assigner on the service account
`confluent_flink_compute_pool`	`isotope-flink-statement-runner`	10 CFU; headroom for 6 INSERTs + ad-hoc SELECTs
`confluent_flink_artifact`	`isotope-flink-udf`	Uploads `ptf/build/libs/isotope-`

Resource	Name	Notes
		`flink-udf.jar`
`confluent_flink_statement` × 20	(see file)	4 ALTER TABLE (event-topic scalar headers) + 3 VIEW (raw + typed produce + typed consume) + 6 sink CREATE TABLE + 1 CREATE FUNCTION (PTF only; UDAF skipped) + 6 INSERT INTO

Useful outputs:

```
terraform -chdir=terraform output environment_id
terraform -chdir=terraform output kafka_bootstrap_servers
terraform -chdir=terraform output schema_registry_url
terraform -chdir=terraform output -raw kafka_api_key     # sensitive
terraform -chdir=terraform output -raw kafka_api_secret  # sensitive
```

4.5.2 Format-by-domain + CCAF UDAF limitation

Format-by-runtime (not -by-domain). CP's reports land on **Avro+SR** (`'value.format' = 'avro-confluent'` in scripts/flink/sql/cp/05_report_sinks.fql). CCAF's reports land on **Protobuf+SR** (`'value.format' = 'proto-registry'` in each sink's WITH clause in terraform/setup-confluent-flink.tf). The two runtimes' SQL is otherwise unshared: CP's lives hardcoded in scripts/flink/sql/cp/, CCAF's lives inline as `confluent_flink_statement` resources in terraform/setup-confluent-flink.tf.

CCAF UDAF limitation. CCAF currently rejects all `CREATE FUNCTION` statements for user-defined aggregate functions ("aggregate functions are not supported"). The `LATENCY_PERCENTILES` UDAF therefore deploys on CP only; the percentile report (`latency_percentiles_flat_1m`) does not exist on CCAF. The other JAR-backed function, `STUCK_TRACE_PTF` (a ProcessTableFunction, not a UDAF), works on both runtimes. The JAR itself is portable — `LatencyPercentilesUDAF` ships with a byte[]-based accumulator and is ready to register when Confluent lifts the restriction. The six CCAF reports are: `latency` (avg/min/max), `topology` (produce-side), `bipartite_topology` (full service↔topic↔service graph), `hop_distribution`, `coverage`, `stuck_trace`.

4.5.3 Sustained traffic — required to see report rows.

Driving traffic — the 4-stage demo against CCAF. App.java reads four optional `-D` properties (`kafka.security.protocol`, `kafka.sasl.mechanism`, `kafka.sasl.jaas.config`, `schema.registry.basic.auth.user.info`) that default to plaintext-no-auth for Minikube. scripts/cc-cli-env.sh pulls the Kafka + Schema-Registry credentials from `terraform output` (both keys are rotated by `module.kafka_api_key_rotation` and `module.sr_api_key_rotation` in terraform/setup-confluent-kafka.tf) and builds the JAAS string.

The thin wrapper scripts/cc-app-run.sh sources the env helper then invokes `./gradlew :app:run` with the six `-D` flags. It accepts two argument styles: **pipeline-position verbs** (`place` / `enrich` / `fulfill` / `ship`) that encode the orders.* topic chain so the 4-terminal demo is one word per terminal, and the **generic `send` / `hop` / `consume` / `sink` passthrough** for ad-hoc inspection on topics outside the demo. Run the script with no args for the full verb list.

```
# Four terminals in pipeline order — same A/B/C/D order as § 4.2. No manual
# env exports — the wrapper sources cc-cli-env.sh, which pulls everything
# from terraform.
scripts/cc-app-run.sh place 'hello'    # A — kick the chain off
```

```
scripts/cc-app-run.sh enrich           # B
scripts/cc-app-run.sh fulfill          # C
scripts/cc-app-run.sh ship             # D — terminal consumer (emits marker)
```

The wrapper hard-fails with a clear message if any of the seven required values is missing, so you'll never silently hand gradle empty `-D` values.

Terminal D prints the same `trace_id` across all three hops, and the CCAF report INSERTs populate as you fire Terminal A — `SELECT * FROM isotope_report_latency_1m`, `SELECT * FROM isotope_report_bipartite_topology_1m`, etc. in the Cloud Console SQL workspace. The bipartite report shows all six edges of the chain (3 produce + 3 consume).

Sustained traffic — required to see report rows. The six INSERT INTO jobs aggregate over `TUMBLE(event_time, INTERVAL '1' MINUTE)` windows, and a tumbling window only emits when the watermark advances past `window_end`. A handful of records bursted from Terminal A within a single 1-minute interval will sit in one open window forever (the most-recent record is the watermark, and it never gets older than itself). Spread traffic across **multiple** windows so the watermark crosses each boundary:

```
# 30 records spaced 5s apart ≈ 2.5 minutes of event-time → spans 3+ windows
for i in {1..30}; do
  scripts/cc-app-run.sh place "burst-$i"
  sleep 5
done
```

Wait ~90 seconds after the *last* record before checking `isotope_report_latency_1m` (and friends) — that's the watermark catching up. The `stuck_trace_alerts_1m` sink only fires for traces that go ≥60s of event time without a fresh hop, so the burst above won't trigger it (every trace gets one record and ends — no stalled in-flight state). To exercise `STUCK_TRACE_PTF`: send a single record to `orders.placed` and don't run the `order-enrichment-service` / `order-fulfillment-service` hops, then keep sending unrelated records elsewhere so the watermark advances past the stuck trace's `event_time + 60s`.

4.5.4 Teardown

```
make cc-flink-reports-down CONFLUENT_API_KEY=$CONFLUENT_API_KEY
CONFLUENT_API_SECRET=$CONFLUENT_API_SECRET
```

Runs `terraform destroy -auto-approve` — deletes every resource above, including the environment itself. Safe to run repeatedly.

5.0 Resources

- Medium Article: Kafka's quiet observability superpower — Kafka Interceptors